

Customer Shopping Behavior/Frequency

Project Overview

This project analyzes the customer shopping behavior or shopping frequency using transactional data having 3900 shopping records across different – different product categories. Our goal is to uncover insight into the data patterns, customer segments, product preferences and subscription status to strategies for business decisions.

Data Summary

- Rows : 3900
- Columns : 18
- Key Features :
 - **Customer Specification:** Age, Gender, Location, Subscription Status.
 - **Purchase Details:** Purchased Items, Category, Purchase Amount, Season, Size, Color.
 - **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type.
 - **Missing Data:** 37 values in Review Rating column.

Exploratory Data Analysis with Help of Python

We began with data preparation and cleaning using Python.

- **Data Loading:** Import data set with the help of Pandas.
- **Initial Exploration:** We use `df.info()` for exploring data structure and use `df.describe()` for summary statistics.

- **Handling Missing Data:** Check for null values and replace those null values with median.
- **Column Standardization:** Rename column(Into Snake Case) name for better readability and documentation.
- **Feature Engineering:**
 - **age_group** column is created by binning customer ages.
 - **Purchase_frequency_days** column is created from purchase data.
- **Checking For Data Consistency:** Check for **discount_applied** and **promo_code_used**, if both are redundant then drop **promo_code_used**.
- **Database Integration:** Connect Python script (which is written on Jupyter Notebook) to PostgreSQL for further analysis.

Analyzing Data Using SQL

We perform a structured analysis in PostgreSQL to answer business questions.

1. Which gender generates more revenue?

Query		Query History	
1	select	gender,	SUM(purchase_amount) as total_revenue
2	from	sales_frequency	
3	group by	gender	

Data Output		Messages	Notifications
	gender	total_revenue	
	text	numeric	
1	Female	75191	
2	Male	157890	

2. Which customers used discount but still spent more than average purchase amount?

The screenshot shows a PostgreSQL query editor interface. The query is as follows:

```
1 select customer_id, purchase_amount
2 from sales_frequency
3 where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from sales_frequency)
4 limit 10
```

The query results are displayed in a table with the following data:

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90

Total rows: 10 of 10 Query complete 00:00:00.287

3. What are top 5 products with highest average review rating?

Sales_Trail/postgres@PostgreSQL 16

Query Query History

```
1 select item_purchased, round(avg(review_rating::numeric),2) as "Average Product Rating"
2 from sales_frequency
3 group by item_purchased
4 order by avg(review_rating) desc
5 limit 8
```

Data Output Messages Notifications

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78
6	T-shirt	3.78
7	Handbag	3.78
8	Belt	3.76

4. Compare the average purchase amount between Standard and Express shipping.

Query		Query History
<pre> 1 select shipping_type, 2 ROUND(AVG(purchase_amount),2) 3 from sales_frequency 4 where shipping_type in ('Standard','Express') 5 group by shipping_type </pre>		
Data Output		Messages Notifications
	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- Do subscribed customers spent more? Compare average average spent and total revenue between subscribers and non-subscribers.

Query		Query History		
1	SELECT	subscription_status,	COUNT(customer_id) AS total_customers,	
2		ROUND(AVG(purchase_amount),2) AS avg_spend,		
3		ROUND(SUM(purchase_amount),2) AS total_revenue		
4	FROM	sales_frequency		
5	GROUP BY	subscription_status		
6	ORDER BY	total_revenue,avg_spend	DESC	

Data Output		Messages			Notifications		
	subscription_status	total_customers	avg_spend	total_revenue			
	text	bigint	numeric	numeric			
1	Yes	1053	59.49	62645.00			
2	No	2847	59.87	170436.00			

6. Which top 5 products have the highest percentage of purchases with applied discount?

8. What are top 3 most purchased products within each category?

Query Query History

```
1 WITH item_counts AS (  
2     SELECT category,  
3            item_purchased,  
4            COUNT(customer_id) AS total_orders,  
5            ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank  
6     FROM sales_frequency  
7     GROUP BY category, item_purchased  
8 )  
9 SELECT item_rank, category, item_purchased, total_orders  
10 FROM item_counts  
11 WHERE item_rank <=3
```

Data Output Messages Notifications

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145

Total rows: 11 of 11 Query complete 00:00:00.295

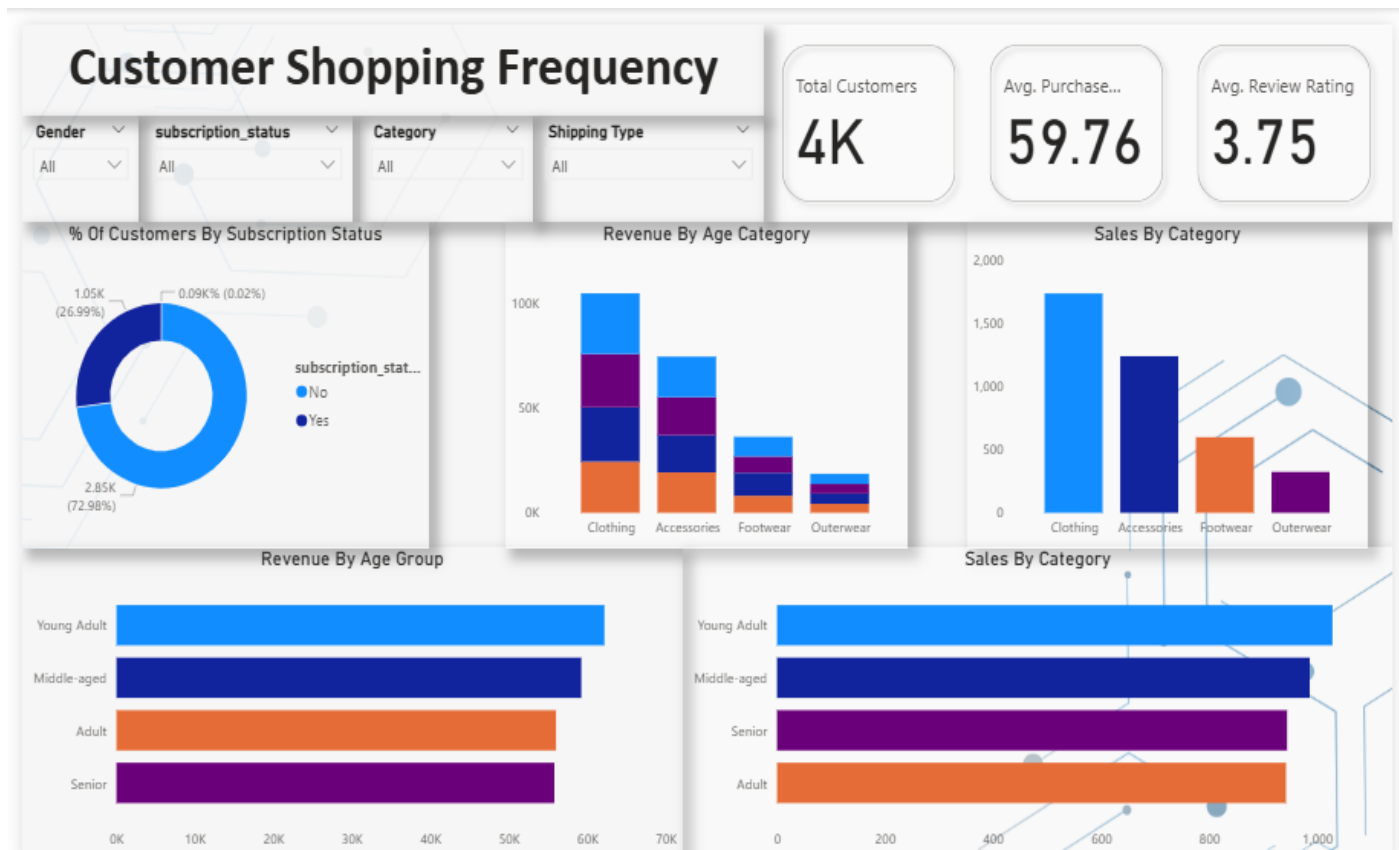
9. Are the customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Query		Query History
<pre> 1 SELECT subscription_status, 2 COUNT(customer_id) AS repeat_buyers 3 FROM sales_frequency 4 WHERE previous_purchases > 5 5 GROUP BY subscription_status </pre>		
Data Output		Messages Notifications
	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. What is the revenue contribution of each age group?

Query		Query History
<pre> 1 SELECT 2 age_group, 3 SUM(purchase_amount) AS total_revenue 4 FROM sales_frequency 5 GROUP BY age_group 6 ORDER BY total_revenue desc </pre>		
Data Output		Messages Notifications
	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Detailed Report Visual



Business Recommendation

- **Subscription Boosting:-** Promote exclusive benefits for new customers for increase in subscribers.
- **Customer Loyalty Program:-** Generate rewards for repeat buyers.
- **Review Discount Policy:-** Balance sales boost with margin control.
- **Product Positioning:-** Best selling and top rated products should be highlighted.
- **Targeted Marketing:-** Focus on high revenue age groups and express shipping users.